juliaTDGL: a guide to the three-dimensional two-band time-dependent Ginzburg–Landau solver

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Abstract

This guide describes what the solver computes and how: the two-band model, its discretisation on a staggered lattice with link variables, the time step, the injection of a transport current, the voltage observable, the convergence of a current ladder, the optional scalar potential and the optional terms (mixed-gradient coupling, heating, Joule heating, bath temperature, charge imbalance), the geometry, and the protocol for reporting a diode efficiency. It ends with how to run the code, what it writes, and what each test checks.

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1 What the solver computes

The solver integrates the time-dependent Ginzburg–Landau (TDGL) equations of a superconductor with two order parameters, Ψ_σ and Ψ_π, coupled by a Josephson term, together with the vector potential $\mathbf{A}$, on a three-dimensional rectangular box surrounded by a buffer, under an applied field and a transport current. It was written to study the geometric superconducting diode: a sample whose shape breaks the mirror symmetry along the direction of vortex motion carries different critical currents in the two directions.

The output of a run is a current–voltage characteristic for both polarities, the vortex content and magnetisation at every current, and the full state at chosen points, from which the critical currents $I_c^{\pm}$ and the efficiency

$$\eta = \frac{I_c^+ - |I_c^-|}{I_c^+ + |I_c^-|} \tag{1}$$

follow.

2 The model

2.1 Free energy and units

The free energy is the two-component Ginzburg–Landau functional with an interband Josephson coupling,

$$\mathcal{F} = \sum_{a=\sigma,\pi} \Big[\alpha_a |\Psi_a|^2 + \frac{\beta_a}{2} |\Psi_a|^4 + \frac{1}{2m_a} \big| (-i\nabla - \mathbf{A})\Psi_a \big|^2 \Big] - \gamma_{12} (\Psi_\sigma^* \Psi_\pi + \text{c.c.}) + \tfrac{1}{2} |\nabla \times \mathbf{A}|^2 . \tag{2}$$

Lengths are measured in the σ-band coherence length ξ_σ, fields in $H_{c2} \equiv H_{c2}^\sigma$, time in τ_{GL}, order parameters in the σ-band equilibrium amplitude at $T = 0$ and currents in the natural unit J_0 of the equations. With this scaling the remaining parameters are

symbol	`Config` field	meaning
κ	`kappa_s`	Ginzburg–Landau parameter, λ/ξ_σ
a_{2n}	`a2n`	equilibrium density of the π band at $T = 0$ (default 0.59)
m_r	`m_ratio`	mass ratio m_π/m_σ (set from `xi_ratio` when given)
R	`xi_ratio`	coherence-length ratio ξ_π/ξ_σ
γ_{12}	`gamma12`	interband Josephson coupling
Γ_r	`gamma_ratio`	relaxation time of π relative to σ
T	`T_red`	reduced temperature
H	`H_const`	applied field along z, in H_{c2}

2.2 Equations of motion

In the gauge $\Phi = 0$ (the default) the dynamics is relaxational:

$$\partial_t \Psi_\sigma = (\nabla - i\mathbf{A})^2 \Psi_\sigma + \left(f\alpha_\sigma - |\Psi_\sigma|^2\right)\Psi_\sigma + \gamma_{12}\Psi_\pi, \tag{3}$$

$$\Gamma_r\, \partial_t \Psi_\pi = \frac{1}{m_r}(\nabla - i\mathbf{A})^2 \Psi_\pi + \left(f\alpha_\pi - |\Psi_\pi|^2\right)\Psi_\pi + \gamma_{12}\Psi_\sigma, \tag{4}$$

$$\partial_t \mathbf{A} = \mathbf{J}_s - \kappa^2 \nabla \times \nabla \times \mathbf{A}, \qquad \mathbf{J}_s = \mathrm{Im}[\Psi_\sigma^*(\nabla - i\mathbf{A})\Psi_\sigma] + \frac{1}{m_r}\mathrm{Im}[\Psi_\pi^*(\nabla - i\mathbf{A})\Psi_\pi], \tag{5}$$

with $\alpha_\sigma = 1 - T$ and $\alpha_\pi = a_{2n} - T$. The normal conductivity is one, which fixes the voltage scale. Γ_r divides the whole right-hand side of the π equation: it sets a relaxation time and does not move any static quantity.

Temperature. By default T enters only the π condensation term and the field equation carries $(1 - T)\mathbf{J}_s$, the convention of the methods paper. Both conventions coincide at $T = 0$. Setting `T_physical=true` gives the equations above, with T felt by both bands and no factor on $\mathbf{J}_s$; the coherence-length ratio is then fixed at its $T = 0$ value. At $T = 0.3$ the uniform equilibrium is $|\Psi_\sigma|^2 = 1 - T$ and $|\Psi_\pi|^2 \simeq a_{2n} - T$.

2.3 Coherence lengths, critical fields and depairing

Linearising about $\Psi = 0$ with the bands decoupled,

$$R = \frac{\xi_\pi}{\xi_\sigma} = \frac{1}{\sqrt{m_r a_{2n}}}, \qquad H_{c2}^\pi = m_r a_{2n} H_{c2} = \frac{H_{c2}}{R^2}. \tag{6}$$

The solver takes R as input and sets $m_r = 1/(a_{2n}R^2)$; it prints the ratio it realises. Raising R lowers the π-band upper critical field as R^{-2}, so a π band with a long coherence length is normal at the fields at which vortices enter: check $\langle|\Psi_\pi|^2\rangle$ before reading any band-resolved number. At finite γ_{12} there is one H_{c2}, the largest root of $(1 - H)(a_{2n} - H/m_r) = \gamma_{12}^2$.

The superfluid stiffness of the pair is $\rho_s = 1 + a_{2n}/m_r = 1 + a_{2n}^2 R^2$ and the effective penetration depth $\lambda_{\mathrm{ef}} = \kappa/\sqrt{\rho_s}$. J_0 is not the depairing current: a uniform superflow gives $j(q) = q[(1 + a_{2n}/m_r) - q^2(1 + 1/m_r^2)]$, with maximum $0.546\, J_0$ at $R = 1.30$ and $2/3\sqrt{3} = 0.385\, J_0$ for one band.

2.4 Geometry as a condensation mask

The sample is always the full box. The geometry enters through a mask $f(\mathbf{r})$ multiplying the condensation coefficients: $f = 1$ in the bulk, $0 < f < 1$ for weakened and $f > 1$ for strengthened condensation. Where $f = 0$ the condensation energy vanishes and the material is locally at its critical temperature: a notch is neither a vacuum cut nor a normal metal, which would need $f < 0$. The local upper critical fields are fH_{c2} for σ and fH_{c2}/R^2 for π: any region with $0 < f < H$ is normal and can shunt the sample (the solver warns at start).

3 Discretisation

3.1 The lattice

The mesh is staggered. Ψ_σ and Ψ_π live on nodes; the vector potential lives on links as the phase $\theta_x = \int A_x\, \mathrm{d}x$ of the link variable $U_x = e^{-i\theta_x}$; the field $\mathbf{h} = \nabla \times \mathbf{A}$ lives on plaquettes. The covariant derivative is $\Psi_{i+1}U_x(i) - \Psi_i$, which is exactly gauge invariant on the lattice. The sample occupies $N_x \times N_y \times N_z$ cells of size $\mathrm{d}x$ inside a box padded by `buf` buffer cells on every face (`bufx`, `bufy`, `bufz` set them per axis). In the buffer the supercurrent vanishes and $\partial_t \mathbf{A} = -\kappa^2 \nabla \times \nabla \times \mathbf{A}$: it is a normal metal of unit conductivity, not vacuum, and carries normal current wherever the electric field is nonzero.

3.2 Boundary conditions

The applied field is imposed on h_z at the outer faces of the buffer; inside, the field is whatever $\nabla \times \mathbf{A}$ gives, so screening, magnetisation and vortex entry are computed rather than prescribed. The order parameters satisfy $\hat{n} \cdot (\nabla - i\mathbf{A})\Psi = 0$ on the six faces of the sample. Each link type has its own mask: a link is live when both of its nodes belong to the sample. A single mask for all three directions would include the low-index transverse rows and exclude the high-index ones, biasing the field equation and the voltmeter towards one edge.

The buffer is a physical parameter, not only a numerical one: the field is clamped at its outer face, inside the region where screening currents live when the padding is thinner than λ. On the notched production sample each doubling of the padding admits one more vortex at zero current. Quote absolute numbers together with the padding.

3.3 Time stepping

Integration is explicit. One substep applies, in this order:

block	operation
[A]	update Ψ_σ, Ψ_π (kinetic, condensation and coupling terms)
[B]	supercurrent on every link, from Ψ and the links of the previous step
[C]	tentative increment of θ: supercurrent minus $\kappa^2 \nabla \times \mathbf{h}$
[C2]	(with `use_phi`) projection by the scalar potential
[E]	links $U = e^{-i\theta}$
[F]	field $\mathbf{h} = \nabla \times \theta$ on plaquettes
[G]	applied field and self-field of the transport current on the boundary
[H]	Neumann condition for Ψ on the sample faces
[I]	advance: the new state becomes the old one (array swaps, no copies)

The time step is the minimum over the diffusion branches of both bands and the field equation,

$$\mathrm{d}t = 0.99 \min\Big[\frac{\eta\,\mathrm{d}l}{4\sqrt{1+\gamma^2}}, \frac{\eta\,\mathrm{d}l\,m_r\Gamma_r}{4\sqrt{1+\gamma^2}}, \frac{\mathrm{d}l}{4\kappa_\sigma^2}, \frac{\mathrm{d}l}{4\kappa_\pi^2}, \frac{\nu\,\mathrm{d}l}{4\zeta^2}\Big], \qquad \mathrm{d}l = \frac{2}{\mathrm{d}x^{-2} + \mathrm{d}y^{-2} + \mathrm{d}z^{-2}}, \tag{7}$$

with $\eta = 5.79$, $\nu = 0.03$, $\zeta = 0.5$, $\gamma = 10$. At $\kappa = 2$ and $\mathrm{d}x = 0.5$ the flat branch binds and $\mathrm{d}t = 4.95 \times 10^{-3}\,\tau_{\mathrm{GL}}$; above $\kappa \simeq 2.9$ the field branch $\mathrm{d}l/4\kappa^2$ takes over, so a run at $\kappa = 25$ takes about 160 times more steps per unit of physical time than one at $\kappa = 2$. Optional diffusive terms add the condition $\mathrm{d}t \le 0.15\,\mathrm{d}x^2/D$. A *cycle* is `lMax`= 40 substeps; convergence is tested once per cycle. This is the scheme of release 1.0.0 and the library default, `Config(:legacy)`.

The production scheme. `Config(:prod)` (release 1.1) removes the three bounds that do not belong to the equations integrated here:

- the term $\kappa^2 \nabla \times \nabla \times \theta$ of block [C] is taken at the new time level (`maxwell_implicit`); the matrix is constant and separable for $\mathrm{d}x = \mathrm{d}y = \mathrm{d}z$, so one solve per step costs a few transforms per axis, and the bound $\mathrm{d}l/4\kappa^2$ disappears;
- the diffusion bounds are those of explicit Euler on diffusion plus reaction, $2/(8/\mathrm{d}l + 2)$ for σ (`cfl_legacy=false`), without the factor $1/\sqrt{1+\gamma^2}$ and the fixed $0.03\,\mathrm{d}l$;
- the π band, whose bound is the smallest at $\xi_\pi/\xi_\sigma = 4$, advances in substeps with σ frozen in the coupling (`pi_substeps=0`, automatic);
- the step is half the bound of σ (`dt_scale=0.5`): at 0.99 of it the stiffest mode changes sign every step and a flowing state at $\kappa = 25$ loses 7 % of its voltage.

It also closes a resistive point once it carries ten crossings per line after `t_min_GL_res` (`conv_by_time`), reads `eps_conv` per cycle of the 1.0.0 step so that the fixed-point criterion is the same in physical time (`eps_conv_legacy`), and uses one BLAS thread. Against the explicit scheme it gives the same onset brackets at $\kappa = 2$ and $\kappa = 25$ and flowing voltages within 0.16 %, at about three times the speed per unit time at $\kappa = 2$ and about a hundred times at $\kappa = 25$. The step cannot grow much further: in strong flow at $\kappa = 25$ the voltage is 0.16 % off at $\mathrm{d}t = 0.03$ and 3 % off at 0.04, because the field and the links are updated once per step during a phase slip. With heat, a bath temperature or drag the π band takes no substeps and only the curl-curl term is implicit, about thirty times faster than the explicit scheme at $\kappa = 25$; heated and dragged flowing and collapsed states at $\kappa = 25$ agree with the explicit ones within 0.25 % in V. The scalar potential runs in the explicit scheme only.

3.4 Symmetry

With $\mathbf{H} \parallel \hat{z}$ and $\mathbf{J} \parallel \hat{x}$ the Lorentz force is along y. Complex conjugation, $\Psi \to \Psi^*$ and $\mathbf{A} \to -\mathbf{A}$, composed with the mirror M_y maps $(+J, +H)$ onto $(-J, +H)$. A geometry invariant under M_y therefore gives $V(-J) = -V(+J)$ and $\eta = 0$ to machine precision; a diode needs a shape that breaks M_y, and an asymmetry along x alone does not. On the lattice the mirror of the interior nodes is $j \mapsto (y_1 + y_2 + 1) - j$. The functions `check_my_symmetry`, `check_mx_symmetry` and `check_rz_symmetry` count the cells that break each symmetry and are printed at the start of every run.

4 Driving and measuring

4.1 Current injection

With $\Phi = 0$, a current imposed as a uniform phase on the links of a contact face is a gauge transformation: it cancels in the lattice curl and in the Neumann condition, and the solver integrates the zero-current problem whatever value is imposed. The solver therefore injects the current through its own field: for a slab of thickness d carrying $\mathbf{J} \parallel \hat{x}$, $h_y = \mp \mathcal{G} J d / 2\kappa^2$ on the two z faces (`inject_mode="selffield"`, gain `inject_gain`$= \mathcal{G}$). This field is the curl itself, cannot be gauged away, and is odd in J. It adds in quadrature to the applied field; keep the total below H_{c2}. The legacy phase clamp (`inject_mode="phase"`) is kept only to reproduce old runs.

4.2 The voltage

The voltage is read from the Josephson relation on the gauge-invariant phase between two measuring planes inset `delta_v` cells from the contact faces,

$$V_\sigma = -\frac{\mathrm{d}}{\mathrm{d}t} \Big\langle \sum_{\text{path}} \arg(\Psi^*_{\sigma,i} U_{ij} \Psi_{\sigma,j}) \Big\rangle_{\text{lines}} . \tag{8}$$

Each transverse line is unwrapped in time separately and only then averaged, so every 2π slip is counted: $V = 2\pi N_{\text{cross}} / (n_{\text{lin}} \Delta t)$. The same construction on Ψ_π gives V_π. The voltage is the slope of a least-squares fit over an averaging window. The plane-to-plane difference of A_x is a static quantity and is not a voltage; it is printed as a diagnostic (`Vest`). Unwrapping is faithful while every line advances by less than π per cycle. The solver records the largest per-line increment of each point (`dmax_arr`), which in developed flux flow is an order of magnitude above the mean rate $|V|\,\mathrm{d}t_{\text{cycle}}$; when it approaches π the remedy is to sample the phase more often (`unwrap_every`), not a smaller time step. A wrapped increment cannot exceed π; the number of increments above 2 rad (`nbig_arr`) bounds the slips that may have been misattributed, which happens when a core passes close to a link. It also reports the voltage on the lines whose whole path has $f = 1$ (`Vf1_arr`), which tells whether slips counted inside a notch, where $\arg(\Psi^* U \Psi)$

is poorly conditioned, contribute to V, and the rate of the line integral of $\mathbf{A}$ over the window (`VA_arr`), which has no core to misattribute but is not gauge invariant: with $\Phi = 0$ the condensate phase between the planes also drifts, and the two voltages differ. For a point that closes as a fixed point, `V_arr` is the slope of the last `V_win` cycles, a short window ($0.1\,\tau_{\mathrm{GL}}$ at $\kappa = 25$); `Vh_arr` and `Vh2_arr` give the slope of the σ and π phases over the second half of every point, which is what the verdict rule needs. `keep_rung_states=true` keeps the state of every closed point in `restart/rungs/`, for analysis after the run.

4.3 Vortex counting and magnetisation

The vortex count is the fluxoid $\oint D = 2\pi n - \Phi_{\mathrm{enc}}$ summed over the plaquettes of the mid-thickness plane, excluding plaquettes where $f \simeq 0$. The mask is the geometry, never the amplitude: masking by $|\Psi|$ removes the cores being counted. The magnetisation is $M = \langle h_z \rangle_{\mathrm{sample}} - H$.

4.4 A current ladder

A run sweeps a list of currents (`J_list`), first $+J$ and then $-J$, each polarity starting from the same zero-current seed. Each rung starts from the converged state of the previous one. A rung closes when either

- the state is a fixed point, $\max|\Delta|\Psi|| <$`eps_conv` per cycle (conv `F`); or
- the state is a limit cycle — flux flow — and the windowed mean of V is stationary (conv `V`),

and in both cases only after a floor of physical time, `t_min_GL` for a static point and `t_min_GL_res` once it is resistive. The floors are in units of τ_{GL}, not of cycles, because the length of a cycle moves with dt. The tolerance `eps_conv` is per cycle; `eps_conv_rate` sets it per unit time instead, so that it does not change with dt. Near onset a flowing state can spend most of each period nearly frozen, and no local tolerance tells it from a fixed point: the verdict rule below does. A rung that reaches `mMax` cycles is marked `*` and is not converged. `Config(:paper)` sets the protocol of the methods paper (tolerance, floors and ceiling).

After each rung the state of that polarity is written to `restart/state_Jpos.jld2` or `state_Jneg.jld2` (compressed JLD2, exact; `checkpoint_format="octave"` writes the text `.mat` of the earlier releases); a run restarted with `restart_pos/restart_neg` continues at the next rung, and `read_state` and `load_checkpoint` open either format, so older runs resume too. Binary MATLAB files need `using MAT` (an optional extension). A file `STOP.txt` in the run folder stops the run after the current rung. `guard_hours` sets a wall-clock budget, and `guard_psi` stops a run whose $|\Psi|$ diverges (usually a time step too large for κ).

5 The scalar potential

With `use_phi=true` the solver leaves the gauge $\Phi = 0$. Each substep computes the tentative link increment δ of block [C], solves

$$D(G\Phi) = D\delta, \tag{9}$$

with G the lattice gradient from nodes to links and D its adjoint, and corrects $\delta' = \delta - G\Phi$, so that $D\delta' = 0$ to machine precision (Coulomb gauge). The order parameters gain $-i\Phi\Psi$ with the time step dt in both bands. $D \circ G$ is the seven-point Laplacian; on the uniform box it is separable and is diagonalised once by three one-dimensional eigenproblems (`phi_method=:separable`, the default), which is an exact direct solve; a sparse LU (`:lu`) gives the same result. With the separable solver a cycle costs 1.55 times a cycle in the gauge $\Phi = 0$.

Φ is a gauge choice: $|\Psi|$, $\mathbf{h}$ and the Josephson voltage are the same with $\Phi = 0$, with Dirichlet and with Neumann conditions on Φ (`phi_bc`); only the split of the voltage between the drop of

Φ, printed as V_Φ, and the drift of the phase changes. Neumann is the natural condition here: there are no electrodes.

Two extensions change the physics:

- **Thomas–Fermi screening** (`lambda_TF`> 0): the operator becomes $D \circ G - \lambda_{\mathrm{TF}}^{-2}$, so the potential is screened over λ_{TF} and the gauge equivalence is broken.
- **Charge imbalance** (`tau_Q`> 0, with Thomas–Fermi): a field Q relaxes towards the projection residual, $\tau_Q\,\mathrm{d}Q/\mathrm{d}t + Q = D\delta'$ (implicit Euler), and the order parameters rotate with $\mu = \Phi + Q$. Without Thomas–Fermi the residual is zero and Q does nothing. With `use_Qss` the residual also diffuses, $\tau_Q(\partial_t Q - D_Q\nabla^2 Q) + Q = D\delta'$.

6 Optional terms

All are off by default. Use one at a time against a control run with the same protocol; with two on, a change cannot be attributed. The coefficients are model values that switch a term on, not material constants.

Mixed-gradient coupling (`use_drag`, `nu_drag`$= \nu$). Adds $\nu\,\mathrm{Re}[(\mathbf{D}\Psi_\sigma)^* \cdot \mathbf{D}\Psi_\pi]$ to the functional, with the corresponding terms in both order-parameter equations and in the supercurrent. The π-side term carries ν/Γ_r and no extra $1/m_r$, which keeps the coupling variational.

Heating (`use_heat`, `D_T`, `eta_J`, `gamma_bath`). A temperature field obeys $\partial_t\theta = D_T\nabla^2\theta + \eta_J s - \gamma(\theta - \theta_{\mathrm{bath}})$, with $\alpha_\sigma = (1-\theta)f$ and $\alpha_\pi = (a_{2n} - \theta)f$ and the source $s = |\partial_t\Psi_\sigma|^2 + \Gamma_r|\partial_t\Psi_\pi|^2$. A rung whose temperature exceeds 0.45 is cut (conv `T`) and ends that polarity.

Joule heating (`use_joule`). Adds the normal-current term $|\mathbf{E}|^2$ to s. Requires heating and $\Phi = 0$.

Contact margin (`heat_margin`). Excludes from s the cells next to each injection face. There the injected normal current converts into supercurrent within about one coherence length, with an electric field and a rotating phase even in the Meissner state; that heat belongs to the contact, not to the sample.

Bath temperature (`T_physical`, `T_red`). See Sec. 2.

7 Geometry

`GEOM_MODE` selects a shape from the built-in catalogue (`SIM`, the clean box; `WEDGE`, a sawtooth notch on the y_1 face; `STAGGER`, `FUNNEL`, `TRI_Y1`, ...); the suffix `_REF` gives the exact M_y mirror, which is the sign control. Any shape can also be read from a file (`GEOM_MODE="CUSTOM"` or `"CUSTOM_REF"`, `geom_file`): a TOML list of regions in units of ξ_σ, painted in order,

```
[[shape]]
type = "triangle"     # circle rect triangle ngon polygon sphere pixels voxels
x = 10.0              # centre; origin at the corner (x1, y1, z1) of the sample
y = 0.0
w = 4.0
h = 3.0
skew = 1.0            # apex offset along the base
rot = 0.0             # degrees about z
z = "all"             # all | bottom | top | buried | [z0, z1]
f = 0.0
```

Because positions are physical, the same file rasterises on any mesh; the catalogue shapes written as files reproduce the catalogue masks bit for bit. The browser editor (`julia julia/editor/server.jl`) draws these files. One picks a face of the sample and paints from it inward, a given number of cells deep and starting a given number of cells inside, so cuts can be buried; figures, gradients of f or of depth, and a palette of condensation strengths complete it. It shows the current, the field and the face through which vortices enter for each sign of J, and checks, while drawing, the mirror symmetry that would force $\eta = 0$, contact with the injection faces and normal shunts.

8 Protocol for reporting an efficiency

A critical current is a bracket: the last silent rung and the first dissipating one of an ascending ladder. A rung integrated for Δt at voltage V carries $|V|\Delta t/2\pi$ vortex crossings per measuring line: below one it is silent, at ten or more dissipating, and in between undecided until it is re-measured on a doubled window. A threshold on the voltage alone, $|V(J) - V(0)| > 10^{-4}$, is weaker: at $\Delta t = 2000\, t_{\mathrm{GL}}$ it reads a thirtieth of a crossing. The run summary and the terminal menu (key `v`) give both brackets with this rule, η at the bracket centres with its range, the undecided rungs, and the voltage non-reciprocity $\sigma_V = (|V_+| - |V_-|)/(|V_+| + |V_-|)$ at the first $|J|$ where both polarities dissipate; a simpler interpolated threshold is printed next to them for comparison. Before reporting η:

1. no region with $0 < f < H$, and $V(J = 0)$ at the floor;
2. the odd part $[V(+J) - V(-J)]/2$ is nonzero (the drive enters); the even part is the rectification;
3. vortices are present somewhere on the ladder;
4. the `_REF` geometry inverts the sign and keeps $|I_c|$;
5. a rigid shift of the defects by $0.05L$ along y leaves the brackets;
6. the asymmetry exceeds that of an M_y-symmetric control run with the same numerics while it dissipates.

The failure modes these checks catch are described in the methods paper: an inert injection, a defect touching a contact, normal shunts, index ranges or mirrors that are not exact, a static drop read as a voltage, averaging before unwrapping, windows too short near the threshold, thresholds set by the current grid, floors counted in cycles, fixed-point tests applied to limit cycles, path dependence of a ladder, a coherence length other than the intended one, an extinct π band still counted, coefficients acting on the wrong quantity, and contact heat counted as sample heat.

9 Numerical modes and reproducibility

`Strict()` routes $|z|$, $e^{-i\phi}$ and the complex product through the C library functions (`hypot`, `sincos`, a fused multiply–add product); it is the mode of the regression tests, which compare against stored states. `Fast()` uses Julia's native arithmetic, vectorises, is about 1.3 times faster and agrees to machine precision. With `use_threads=true` the planes in z are updated in parallel; each plane writes distinct cells, so the result does not depend on the number of threads. At the start of each run a self-test compares the vectorised kernel with a reference triple loop.

On one core of a laptop (Intel i5-1235U) a cycle of the $40 \times 24 \times 6$ production box costs 0.030 s, so a rung on a $1500\, \tau_{\mathrm{GL}}$ floor takes about four minutes; with the scalar potential, 0.046 s.

10 Running

10.1 From Julia

```
using Pkg; Pkg.activate("julia")
using TDGL
sde = run_case(Config(GEOM_MODE="WEDGE", Nx_sc=40, Ny_sc=24, Nz_sc=6,
                      dx=0.5, dy=0.5, dz=0.5, kappa_s=2.0, xi_ratio=1.30,
                      H_const=0.50, J_list=[0.040, 0.045],
                      t_min_GL=1500.0, t_min_GL_res=2000.0, eps_conv=1e-5,
                      out_dir="runs/first", run_tag="first"))
brackets(sde)        # Ic+- as brackets with the verdict rule, eta and its range
```

The same, smaller and in about a minute, is `examples/first_run.jl`.

10.2 From the command line and the terminal menu

`julia/scripts/run_preset.jl` launches one run with flags for the geometry (`-geom`, `-shape`), the parameters (`-kappa`, `-H`, `-xi-ratio`, `-gamma12`, `-T`), the currents (`-J`), the mesh (`-mesh` or `-Nx -Ny -Nz -dx`), the times (`-window` or `-tmin -tminres`), the scalar potential (`-phi off|coulomb|tf`) and one optional term (`-drag`, `-heat`, `-qss`), and the scheme (`-scheme prod|legacy`; prod unless Φ or Q needs legacy; with heat, drag or T, prod only where its step is at least twice the explicit one; a resumed run keeps its scheme). `julia julia/scripts/tdgl_tui.jl` builds the same command step by step, lists the runs, stops them cleanly and resumes them.

10.3 Where runs are written

`run_case` writes to `out_dir`, any path, created if needed; `run_preset.jl` writes to `julia/runs/<name>/` unless `-dir` says otherwise. The terminal menu lists every run below `julia/runs`, at any depth (`julia/runs/project/runs/case` is found), and every `runs/` folder one or two levels below the root of the repository (`project/runs`, `group/project/runs`); a run kept anywhere else is not scanned. Runs started outside the menu (a script, a cluster job) appear as read-only rows, with their progress read from their own output.

10.4 On a cluster

The menu is for a workstation; on a cluster a run is one batch job with one core (`JULIA_NUM_THREADS=1`, `OPENBLAS_NUM_THREADS=1`: more BLAS threads only oversubscribe). Ladders at different fields, shapes or parameters do not talk to each other, so the natural use of a cluster is a job array, one task per run: N tasks give N times the throughput with no change to the solver. `examples/slurm_array.sh` runs the notched box at five fields:

```
#SBATCH --array=0-4 --cpus-per-task=1 --mem=2G --time=24:00:00
export JULIA_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1
FIELDS=(0.015 0.03 0.06 0.09 0.12)
H=${FIELDS[$SLURM_ARRAY_TASK_ID]}
julia --project=julia julia/scripts/run_preset.jl --geom WEDGE --kappa 25 \
      --xi-ratio 4 --H $H --J 0.02,0.04,0.06,0.08,0.10,0.20,0.30,0.40 \
      --mesh production --window long --dir runs/field_H$H ${RESUME:+--restart}
```

A task stopped by the time limit resumes from its last closed point when it is submitted again with `RESUME=1`: the folder name must not depend on the job number. `examples/slurm_job.sh` is the same for a single run. Results are read without the menu: `julia julia/scripts/tdgl_tui.jl -results`, or `brackets(sde)` in Julia.

10.5 What a run writes

file	content
`job.json`	the parameters of the launch
`j_history.csv`	one line per closed rung: J, V, class, vortices
`run_info.txt`	time step and cycle length, to count crossings
`output/RUN_SUMMARY_*.txt`	parameters, $I_c^{\pm}$, η and a convergence table
`restart/state_J{pos,neg}.jld2`	full state (compressed JLD2; `.mat` with `checkpoint_format="octave"`)
`output/snapshots/`	mid-plane $\lvert\Psi\rvert^2$ as PGM and CSV
`output/vtk/`	legacy VTK for ParaView (`vtk_every`)

Any checkpoint converts to VTK with `julia/scripts/vtk_from_checkpoint.jl`.

11 Tests

`julia -project=julia julia/test/run.jl` runs about 56 000 assertions in a minute and a half. They check the kernel against the reference loop and across threads; regression against two stored states (a Meissner state and a state with vortex flow); the lattice Landau level (1.6 % low at $\mathrm{d}x = 0.5$, $H = 0.5$) and the surface-nucleation ratio H_{c3}/H_{c2} (0.05 % from 1.69461); that Γ_r moves no static quantity; that Φ is a gauge freedom, also with a Thomas–Fermi length, as long as the residual does not relax; that the separable solver equals the LU; every optional term and its guards; shapes read from files; and the VTK writer.

12 Limitations

- Standard TDGL with a scalar relaxation time per band; no generalised TDGL, no mass tensor, no microscopic derivation of Γ_r.
- The two-component functional has two parameters, m_r and a_{2n}, for three targets of a material such as MgB_2 (coherence-length ratio, stiffness partition and gap ratio); not all can be matched at once.
- A notch ($f = 0$) is locally at T_c: neither vacuum nor a normal metal.
- The buffer is a normal metal and a systematic: absolute thresholds depend on its width. The box has no common electrodes at its ends, so a thick buffer carries normal current in parallel with a silent sample (with 24 cells around the $40 \times 24 \times 6$ box, 62 % of it); one cell keeps that path small. A thicker buffer is not a closer model of a film in vacuum.
- The explicit scheme costs time steps quadratically in κ; the production scheme removes it except with the scalar potential.
- The silent state is driven, not an equilibrium. The current enters through the contact faces as normal current and turns into supercurrent within one or two ξ_σ; those layers carry a steady electric field. The current at which the silent state starts to flow therefore depends on the relaxation rates and on the contact, not only on the free energy.